AI Resume Analyzer and Job Recommendation System

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Batch: Artificial-Intelligence-August-B-2026-batch

Project Objective

Develop an NLP-based application that that helps students understand how well their resume matches selected job roles. The system should provide a match score, list important missing skills, and generate a simple learning roadmap.

Methodology

Step	Implementation
Resume Upload	Upload a resume in PDF or DOCX format.
Text Extraction	Extract text from the uploaded resume using pypdf or python-docx.
Text Cleaning	Convert the text to lowercase and remove unnecessary characters and extra spaces.
Skill Extraction	Identify technical skills using a predefined skill dictionary and keyword matching.
Job Matching	Use TF-IDF and cosine similarity to compare the resume with job-role requirements.
Recommendations	Rank job roles based on match scores and recommend the top three roles.
Skill Gap	Compare the skills with the required skills of the selected role and identify missing skills.
Learning Roadmap	Provide basic learning steps for the identified missing skills.

Technologies Used

Python, Pandas, Scikit-learn, pypdf, python-docx, and Streamlit.

Results

The system successfully analyzes PDF and DOCX resumes and provides technical skills, match scores for five job roles, the top three recommended roles, missing skills for a selected role, and a basic learning roadmap. The application was also tested for resume extraction, skill extraction, job matching, skill-gap analysis, and a no-skill resume case.

Limitations

The current system uses a predefined skill dictionary and five predefined job roles.

Conclusion

This project helped me understand how **NLP can be applied to a real-world problem** such as resume analysis and job recommendation. It provided practical understanding of extracting and cleaning unstructured text, identifying skills using keyword matching, and using **TF-IDF and cosine similarity** for job-role matching. The project also helped me understand how different components of an AI application can be connected from **data processing and analysis to a working Streamlit application**.